

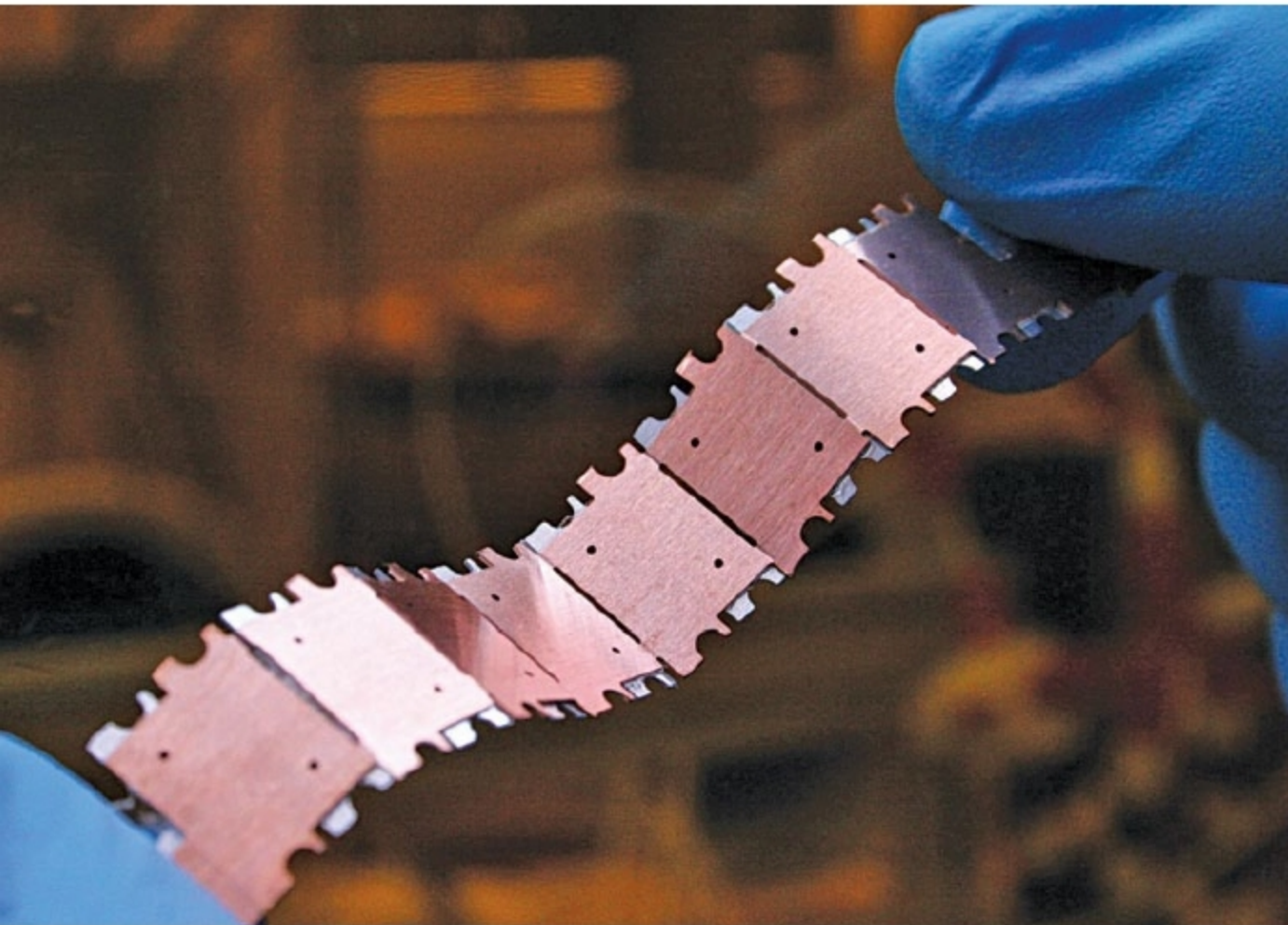
Pliable micro-batteries for wearables

Wearables are portable systems that contain sensors to collect measurement data from human bodies. There are technical considerations when it comes to powering these smart sys-

tems, such as durability and energy density. Also, there are material requirements such as weight, flexibility and size, in addition to these being successfully combined. To solve this issue, experts at Fraunhofer Institute for Reliability and Microintegration IZM, Germany, have developed a prototype for a smart wristband that collects data first hand.

The most remarkable feature of the silicone band is its three gleaming green batteries. Boasting a capacity of 300mAh, these batteries can store energy of 1.1W and lose less than three per cent charging capacity per year. With these parameters, the prototype has a much higher capacity than smart bands currently available at the market. This enables the band to supply even demanding portable electronics with energy.

Instead of making the batteries extremely pliable at the cost of energy density and reliability, researchers designed tiny, powerful batteries and optimised mounting technology. The batteries are pliable in between segments. Hence, the smart band is flexible while retaining a lot more power than other smart wristbands.



Mechanically-flexible micro-battery strip made from segmented battery cells (Credit: Fraunhofer IZM)

Biodegradable electronic implant that speeds up nerve regeneration

Researchers at Northwestern University, USA, and Washington University School of Medicine in St. Louis, USA, have collaboratively developed the first bioresorbable electronic medicine—an implantable, biodegradable wireless device that speeds nerve regeneration and improves the healing of a damaged nerve.

During testing, the device delivered regular pulses of electricity to damaged peripheral nerves in rats after a surgical repair process, accelerating the regrowth of nerves in their legs and enhancing the recovery of muscle strength and control.

The wireless device is the size of a dime and the thickness of a sheet of paper. It operates for about two weeks before naturally absorbing into the body.

The scientists envision that such transient engineered technologies could one day complement or replace pharmaceuti-



The engineered system provides active, therapeutic function in a programmable, dosed manner and then naturally disappears into the body, without a trace